

CRANBERRY: An RNA Dynamics Model with Sugar Puckering and Noncanonical Base Pairing

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We introduce a new coarse-grained model “CRANBERRY” that incorporates sugar puckering and non-canonical base pairing, two factors central to RNA structure and dynamics, yet rarely included in most coarse-grained models. Our model is parameterized through a contrastive divergence approach, combined with fine-tuning strategies to improve accuracy in generating disordered states, a feature that is critical for the accurate description of thermodynamics. This two-stage training procedure greatly enhances cooperative folding behavior. Due to these advances, the model’s predictive performance is comparable to that of all-atom force fields for native-state structural fluctuations. Furthermore, CRANBERRY exhibits better agreement with experimental data on stacking free energies and disordered structures measured by Small Angle X-ray Scattering. In addition, CRANBERRY can reversibly fold tetraloops with an RMSD_{min} of 1.4 Å *de novo*, which continues to be challenging for all-atom models. It predicts melting temperatures in agreement with experimental values, and with a greater cooperativity than all-atom predictions.

RNA folding | molecular dynamics simulations | force field

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Introduction

The structure of RNA affects a wide range of processes, from biological function to therapeutic targeting, providing information on gene (1), protein translation (2, 3), and catalysis (4, 5). The static perspective of the RNA structure has evolved into an ensemble-based view of the dynamics of RNA (6–11). The dynamic aspect of RNA is likely critical for cellular functions (1, 11) and human diseases (12). RNAs exhibit rugged folding free energy landscapes (FES) (7, 13–16). This opens up the possibility that the biologically relevant conformations are not always their ground states. These observations help motivate the development of accurate and efficient models for generating RNA structural ensembles and the underlying structural transition pathways, the central goal of this work. Our aim is not to compare our model directly with models designed primarily for structure refinement or prediction, such as deep-learning models (17–21), or

certain Monte Carlo (MC) models (22–26). These models are effective for structure prediction, but generally not well suited for modeling RNA structural dynamics and folding pathways (27–29).

Several all-atom force fields for RNA have been developed (30, 31). A widely used model is OL3 (32), which offers good performance near the native basins of various RNAs. However, biologically relevant RNA dynamics on the microsecond time scale and beyond is difficult to capture with all-atom models due to computational limitations (30, 33), even for systems as small as tetraloops (34–39). Additionally, RNA force fields (31, 40–44) such as OL3 (32) or DES-AMBER (33) face challenges in correctly predicting disordered states and thermodynamics while maintaining the stabilities of the native structure. This apparent trade-off among native states, disordered states, and thermodynamics suggests an underlying imbalance between various energy terms (36, 43).

To accelerate computation while systematically addressing the imbalance issue, we choose to develop a coarse-grained (CG) model that accelerates sampling of the RNA dynamics by reducing the number of degrees of freedom and smoothing free energy landscapes. (30, 45) The increased speed also makes it practical to refine the model iteratively.

Within CG RNA models, two parameterization strategies dominate (30, 45–47). Knowledge-based (KB) models derive statistical potentials from structural databases via reference-state formalisms (48). BRIQ (23, 24), SPQR (25, 26), SimRNA (22), isRNA (49, 50), RACER (51–53), and HiRE-RNA (54, 55) are examples of this type. Reference-state methods face a self-consistency problem and require iterative parameterizations, which are computationally expensive (49, 56). Most CG models of this type aim only to predict or refine structures (30, 45), with notable exceptions such as RACER (51–53) and Hi-RE RNA (54, 55) that can fold RNA motifs *de novo*. Thermodynamic quantities can be difficult to define rigorously.

The second approach, i.e., models based on thermodynamics,

usually starts with nearest-neighbor (NN) parameters (57) and aims to match melting temperatures and other thermodynamic data. oxRNA (58), TIS (5, 59, 60), Martini (61), and 3SPN.2 (62) belong to this category of models. They capture thermodynamics well, but typically require prior knowledge of RNA structures, rarely produce *de novo* folding dynamics, and often assume A-form helices with limited support for noncanonical base pairs (6, 7).

RACER, originally knowledge-based, matches unfolding free energies but shows reduced native stabilization after thermodynamic optimization, echoing all-atom trade-offs (43). To the best of our knowledge, it remains the only CG model that aims to recover both thermodynamics and *de novo* folding across diverse RNAs, thus serving as our benchmark.

In summary, KB and thermodynamics-based models face four main challenges: 1) costly/ill-posed reference states (KB); 2) limited noncanonical pairing (thermo); 3) difficulty achieving both thermodynamics and folding; 4) under-represented sugar-puckering dynamics that influence binding, catalysis, and primer extension (63–66). (SPQR represents sugar puckering, but by discrete MC moves.) To address these challenges, we develop CRANBERRY (Coarse-grained RNA model with ANisotropic BasE interactions and Ribose Ring puckering dYnamics): 1) We use the contrastive divergence (ConDiv) method (67–70), which has proven effective for parameterizing the CG protein model *Upside* (71). This trajectory-based iterative method does not require a postulated reference state and is computationally less expensive than rigorous reference state approaches. 2) We further parameterize the model by fine-tuning against stacking free energies and disordered RNA structures. This two-step training is analogous to the dual-target training used in our subsequent work to improve *Upside* (72). 3) We design energy functions for non-canonical base pairings that avoid degeneracy and have a clear correspondence to all-atom base-pairing geometries. 4) We develop a many-body bonded potential to represent the collective transition between the two dominant sugar-puckering states. Overall, CRANBERRY achieves performance comparable to all-atom force fields for native fluctuations, and outperforms all-atom force fields in disordered states and thermodynamics. CRANBERRY also outperforms the CG model RACER in these three benchmarks. Additionally, CRANBERRY predicts melting temperatures within 15 K of their experimental values and shows greater cooperativities (sharper melting curves) than all-atom force fields. Moreover, CRANBERRY can fold the tetraloop ggcGCAAgcc into native conformations *de novo*, reversibly, with an RMSD_{min} of 1.4 Å. CRANBERRY also reproduces the destabilization of the C2'-endo state in an A-form helix.

Results and Discussion

We first introduce CRANBERRY's representation and then discuss two important aspects of our energy function design: sugar puckering motion and anisotropic base pairing. Then we evaluated the performance of CRANBERRY and compared it with OL3, DES-AMBER, and RACER. The models were tested for their ability to fold or stabilize native struc-

tures, describe disordered states, and predict stacking free energies. We further tested CRANBERRY's ability to reproduce the melting temperatures compared with those of DES-AMBER and *reversible and de novo* folding of tetraloop ggcGCAAgcc. We demonstrated the ability to predict sugar puckering FES for one nucleotide or a double helix using CRANBERRY.

Model. In CRANBERRY, each nucleotide is represented by 6 sites: one for the phosphate group, two for the sugar, and three for the base (Fig. 1A). Compared to isRNA2 (50) or RACER (51–53), we use two sites rather than one, C3' (S3) and C2' (S2), to represent the sugar. These two sites allow us to model the sugar pucker transition between C3'-endo and C2'-endo, as discussed in the next Section **Results and Discussion: Sugar Puckering Motion**. The base mapping is the same as in RACER (51–53) (Fig. 1C). We note that the mapping for the base does not have a significant impact on the model energy function as long as three sites are chosen, because the base stacking and base pairing interactions only take the center of geometry and the normal vector of the bases as input, as discussed in Section **Methods: Nonbonded Interactions**. Following RACER, placing the sites near the base edges better captures the excluded volume of the bases. We put a virtual site at each base's center of geometry, termed BC (base center), and another virtual site, termed BN (base normal), such that the vector from BC to BN defines the base-plane normal (Fig. 1B). The normal vector is chosen to point towards the 5' direction in a canonical A-form helix. Both BC and BN are used in base stacking and pairing potentials to define base-plane orientation. Additionally, BC is used in steric calculations to better represent the excluded volume.

Sugar Puckering Motion. Sugar puckering refers to the out-of-plane motion of the five-membered ribose ring (73–77). This collective motion couples the degrees of freedom of the backbone to the base (78, 79). RNA ribose most frequently adopts two conformations: C3'-endo and C2'-endo. This motion is well described by the phase angle (77, 80, 81), a collective variable derived from the coordinates of the five ring atoms. The Cremer-Pople definition (80) of the phase angle projects all five ribose sites to the mean plane of the ribose (Fig. 2B), and then calculates the phase angle from the projection distance z_j for each site j

$$\begin{aligned} a &= \sqrt{2/5} \sum_{j=1}^5 z_j \cos(4\pi(j-1)/5), \\ b &= \sqrt{2/5} \sum_{j=1}^5 z_j \sin(4\pi(j-1)/5), \\ P &= \arctan(b/a). \end{aligned} \quad (1)$$

The phase angle can distinguish all possible conformations of the ribose, including C3'-endo (phase angle in the range $0 - 36^\circ$) from C2'-endo ($157^\circ - 180^\circ$) (Fig. 2A). Similarly, north and south conformations are defined as conformations in the north ($0^\circ < P < 90^\circ$ or $270^\circ < P < 360^\circ$) or the south

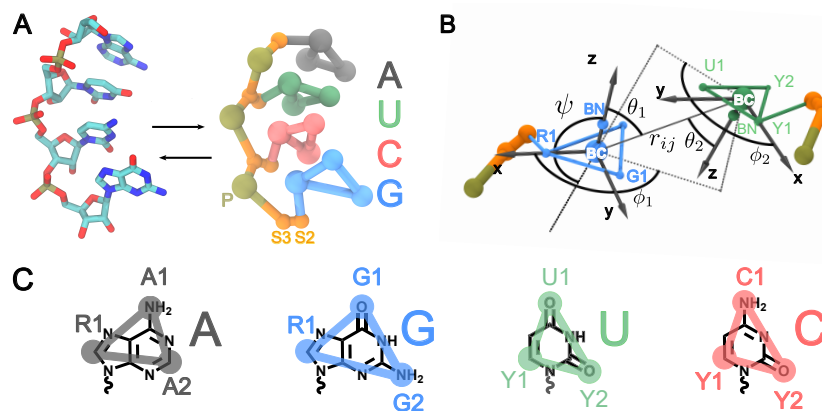


Fig. 1. The CRANBERRY model. (A) Mapping between all-atom and coarse-grained resolutions, with coarse-grained sites placed on the phosphate, C3' and C2' of the sugar, and 3 atoms of the base (B) Positions of virtual sites BC (base center) and BN (base normal), and definitions of degrees of freedom r_{ij} , θ_1 , θ_2 , ϕ_1 , ϕ_2 , ψ between two interacting bases (G-U in the figure). BC is placed at the base center of geometry, and BN is positioned so that the vector from BC to BN defines the base normal. The direction is chosen such that BC-BN points towards the 5' direction in an A-form helix. r_{ij} is the distance between base i and j . We define a local coordinate system for each base, aligning the z-axis with the base normal and the x-axis with the vector BC-R1/Y1. θ_1 , θ_2 are angles between $\vec{r}_{ij}$ and the z-axis of each base. To calculate ϕ_1 , ϕ_2 , we first project the vector $\vec{r}_{ij}$ onto each base plane, then calculate the angle between the projected vector and the x-axis of each base. Note that θ_1 , θ_2 range from $[0, 2\pi]$. r_{ij} , θ_1 , ϕ_1 is the coordinate of BC of base j in the local spherical coordinate system centered at BC of base i . (Mathematical definitions can be found at Eq. 8) (C) Mapping of the four base types, overlaid with their chemical structures.

($90^\circ < P < 270^\circ$) part of the phase angle plot (82). However, the phase angle calculation requires all five ring atom coordinates. This raises the question of how to model this collective transition and distinguish between C3'-endo and C2'-endo in a coarse-grained model. We address this by developing a linear regression model to predict the phase angle from the coarse-grained sites and use it to interpolate between two sets of bonded energy functions corresponding to the C3'-endo and C2'-endo states (Fig. 2E).

The predicted phase angle P_j is defined by

$$P_j = \sum_{i \in \text{bonds}_{\text{sp}}} w_i^r r_i + \sum_{i \in \text{angles}_{\text{sp}}} w_i^\theta \theta_i + \sum_{i \in \text{dihedrals}_{\text{sp}}} \left[w_{is}^\phi \sin(\phi_i) + w_{ic}^\phi \cos(\phi_i) \right] + b_0, \quad (2)$$

which is a linear combination of coarse-grained bonds, angles, and the sine and cosine of dihedrals. We denote the bonds, angles, and dihedrals that is highly correlated with sugar pucker as $\text{bonds}_{\text{sp}}^j$, $\text{angles}_{\text{sp}}^j$, $\text{dihedrals}_{\text{sp}}^j$ with j denoting the nucleotide index, and “sp” indicating the relevance of sugar pucker. For example, the coarse-grained angle P-S3-S2 is highly correlated with the sugar pucker state (Fig. 2D). The weights are obtained by linear regression on the fragment data set (Materials and Methods: Data Set). 10% of the fragment data set is used as the test set, and the R^2 is reported on the test set. We used the sine and cosine of the dihedral angles, rather than the dihedral angle itself, because this ensures a smooth transition in the predicted phase angle when one of the dihedral angles crosses from -180° to 180° . As it turns out, the linear regression model works well in this case with $R^2 = 0.92$. The choice of coarse-grained sugar sites also matters (Fig 2C), and we used the optimal mapping, C3' and C2', which gives the highest R^2 for the phase angle prediction. These two sites likely best differentiate C3'-endo and C2'-endo conformations. The parameters

related to the predicted phase angle P_j are listed in Table S7 of the Supporting Information (SI) Appendix. We use this predicted phase angle to interpolate between bonded energy functions parameterized individually for C3'-endo and C2'-endo (Fig. 2E), with definitions shown in Eq. 4 in Section Materials and Methods: Sugar Puckering Energy Function.

Base Pairing Geometries. All three edges of a base can participate in base pairings, which are termed Watson-Crick (WC) edge, Hoogsteen edge, and sugar edge (83–85) (Fig. 3A). To differentiate different edges, we propose to use the ϕ_1 and ϕ_2 angles, which are the angles between the center of mass vectors projected to the base plane and the x-axis of the local coordinate systems (Fig. 1B, defined in Eq 8). In a canonical GC WC-WC base pairing, ϕ_1 and ϕ_2 are both around 180° (Fig. 3b). In a sheared GA pair (Sugar-Hoogsteen), $\phi_1 \approx 290^\circ$ and $\phi_2 \approx 60^\circ$ (Fig. 3c). As indicated by these two examples, ϕ_1 and ϕ_2 are good indicators of base pairing geometry (Fig. 3A, ϕ_1 and ϕ_2 for all the base pairs in the fragment data set (Materials and Methods: Data Set), colored by the accurate assignment of base geometry performed using DSSR (86–88)). Different geometries are clearly well-separated in the scatter plot $\phi_1 - \phi_2$. Our definition of $r, \theta_1, \theta_2, \phi_1, \phi_2, \psi$ is equivalent to that of the SPQR model (25). In the SPQR model (25), the 6D joint distribution of two bases ($r, \theta_1, \theta_2, \phi_1, \phi_2$) is simplified by assuming that the base orientations are independent given a fixed inter-base distance. In contrast, we retain the key correlation between the base orientations between ϕ_1 and ϕ_2 by decomposing different base pair geometries. This decomposition is required because of the different underlying all-atom hydrogen-bonding patterns. Taking into account base types, edge types, and base orientations, there are a total of $2 \times \left(\binom{12}{1} + \binom{12}{2} \right) = 156$ distinct base pairing geometries. We model only the 25 most frequent base pairing geometries found in the fragment data set, accounting for 88% of the

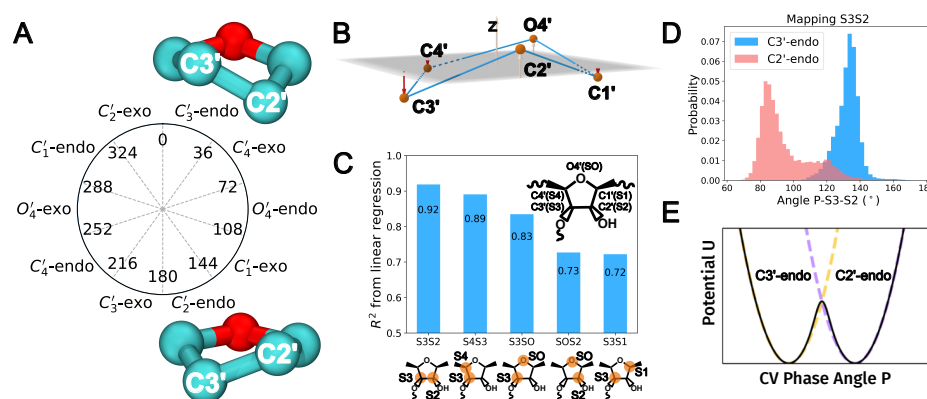


Fig. 2. Schematics and results for sugar pucker energy function design. (A) Possible conformations of ribose and the corresponding phase angles, with C3'-endo and C2'-endo visualized. (B) Schematics for the Cremler-Pople method for calculating the phase angle. The coordinates of each atom in the ribose are projected to the mean plane, and the length of the projection is used to calculate the phase angle. (C) R^2 of the test set from the linear regression if different mapping schemes are chosen. (D) Example distribution of coarse-grained angle P-S3-S2 with respect to sugar pucker states. (E) A schematic representation of the sugar pucker bonded energy function, which can be thought of as a collective variable that provides smooth interpolation between C3'-endo energy functions and C2'-endo energy functions.

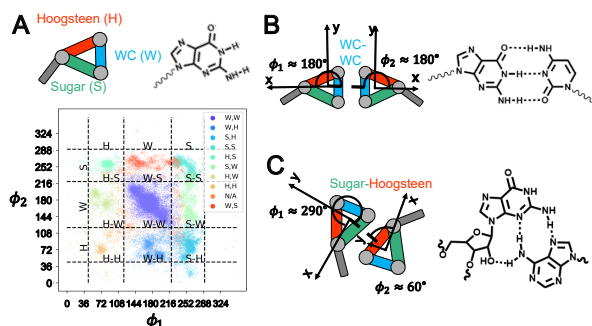


Fig. 3. Schematic representations of base interaction edges and ϕ_1 - ϕ_2 plot of base pairing. (A) Schematic of the three base interaction edges (Watson-Crick, Hoogsteen, and Sugar) (83–85) and the chemical structure of Guanosine. The ϕ_1 - ϕ_2 (defined in Fig. 1B and Eq. 8) plot illustrates a clear separation between different base geometries. The colors of the scatters are determined by DSSR (87). (B) Example of a WC-WC interaction in a GC pair, where both ϕ_1 and ϕ_2 are near 180° , along with the corresponding chemical structure. (C) Example of a Sugar-Hoogsteen interaction, where $\phi_1 \approx 290^\circ$ and $\phi_2 \approx 60^\circ$, shown with its corresponding chemical structure.

base pairs. and leave the rest rare base pairings as zero energies. Definitions can be found in Eq. 8 in Section **Materials and Methods: Nonbonded Interactions**, where an anisotropic energy function based on ϕ_1, ϕ_2 is developed for base pairings.

After designing the important interactions in CRANBERRY, the parameters are trained and optimized according to the procedures described in Section **Materials and Methods: Parameterization Approach**. To test the ability of CRANBERRY, we benchmark the model in the following sections on native and disordered structures, stacking, and melting thermodynamics, *de novo* folding, and sugar pucker.

Structural stability. CRANBERRY generally stabilizes the motifs (tetraloops 1ZIH, 2KOC, double helix 3ND4, and pseudoknot 1L2X) better than RACER (Fig. 4 and Fig. S4 of the SI Appendix). For RACER, the pseudoknot 1L2X melts at 300 K, whereas CRANBERRY maintains its structure with an RMSD = $4 \sim 6$ Å of the native structure. CRANBERRY can stabilize the two tetraloops to an RMSD of $1.5 \sim 5$ Å. For

the A-form helix 3ND4, CRANBERRY produces a more accurate structure than RACER, which exhibits a broad peak from 2.5 Å to 10 Å. For the two tetraloops, RACER predicts a rapid equilibrium between native-like structures and unfolded structures. This destabilizing behavior of RACER could potentially come from either the lack of modeling for most non-canonical base pairings or over-destabilization of the structure to match thermodynamic data. This underscores the challenge of balancing structural stability with thermodynamic accuracy.

The structural distributions of DES-AMBER and OL3 are largely concordant. DES-AMBER yields slightly higher RMSDs for 1ZIH and 1L2X than OL3, which is consistent with reports that DES-AMBER allows more natural fluctuations due to its enhanced modeling of disordered RNA and thermodynamics (33, 43). For OL3 and DES-AMBER, the RMSD is generally 1–2 Å lower than CRANBERRY. This discrepancy may have multiple origins. First, CRANBERRY does not model all base-pairing interactions and is coarse-grained, which makes it, in general, less accurate in capturing detailed interactions. However, there is also the possibility that OL3 and DES-AMBER over-stabilize these native structures. This hypothesis is partially supported by the performance of OL3 in disordered RNA, where it produces an overcollapsed state for rU30 (Fig. 5C) and rU40 (33) with a radius of gyration (R_g) 10 Å below experimental measurements (Fig. 5C). This is also supported by later benchmarks on stacking free energies as shown in Fig. 6 and has been reported previously (40, 41). This suggests that OL3/DES-AMBER might be too “sticky,” similar to what has been observed for all-atom protein force fields (89–91). The actual native fluctuations may lie between the current estimates from all-atom force fields and CRANBERRY.

SAXS measurements of disordered RNAs: rU30 and rA30. CRANBERRY’s simulations produce SAXS (Small Angle X-ray Scattering) data that are very similar to the experimental measurements on rU30 (Fig. 5A and B). CRANBERRY’s ability to simultaneously obtain native structures

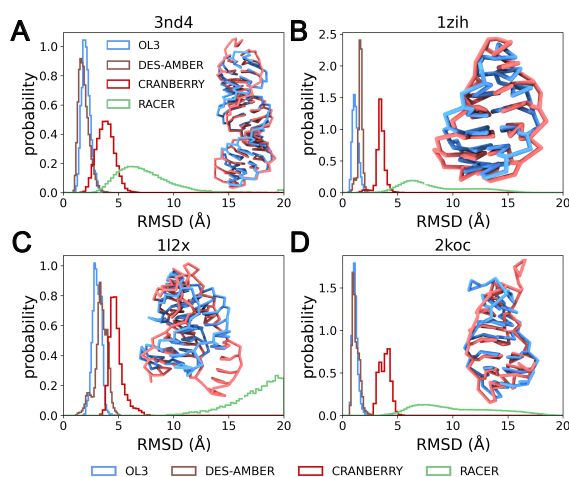


Fig. 4. Native simulations for (A) a canonical A-form double helix 3ND4 (B) a tetraloop 1ZIH (C) a pseudoknot 1L2X (D) a tetraloop 2KOC by OL3(32), DES-AMBER (33), CRANBERRY, and RACER (53). In each plot, the blue structure is the crystal structure, and the red structure is the snapshot of the last frame of the CRANBERRY simulation.

and unfolded conformations (i.e., a very high degree of all-or-none cooperativity) implies that our training procedure captures the delicate balance between various potential energies (interactions of the backbone side chain, base stacking, electrostatic repulsions, sugar puckering bonded interactions, and internal stiffness of the chain).

OL3 produces a collapsed conformation for the disordered rU30 with $R_g \sim 10 \text{ Å}$ below the experiments (Fig. 5A and F), which is similar to the behavior of rU40 as shown in Ref. 33. DES-AMBER performs better, with R_g values that are slightly higher than the experiments. For this benchmark, we use DES-AMBER 3.20 (92) instead of DES-AMBER 2017 (33), because version 3.20 improves phosphate solvation and therefore provides more accurate predictions for disordered RNAs, making it a more appropriate benchmark. In contrast, DES-AMBER 2017 tends to exhibit transient hairpin formation (33) and produces R_g values that are 4–5 Å smaller than the experimental measurements (SI Appendix Fig. S5). For the remainder of this paper, however, we continue to use DES-AMBER 2017, since the melting temperature calculations are computationally expensive and are only available for DES-AMBER 2017. For RACER, the R_g value matches the experiment at high salt, but exhibits higher salt dependency compared to experiments at low to intermediate salt regimes (Fig. 5A and C).

Regarding the scattering profile of rU30 under 200 mM NaCl, CRANBERRY shows excellent agreement with the experimental results (Fig. 5B). In fact, CRANBERRY even predicts good scattering profiles for rU30 across the whole range of salt concentrations (SI Appendix Fig. S6). DES-AMBER exhibits slightly higher slopes in the high q range compared to the experiments, suggesting the simulated rU30 might be slightly too rigid at short length scales along its backbone. This may stem from the over-stabilization of UU stacking observed in DES-AMBER (Fig. 6A).

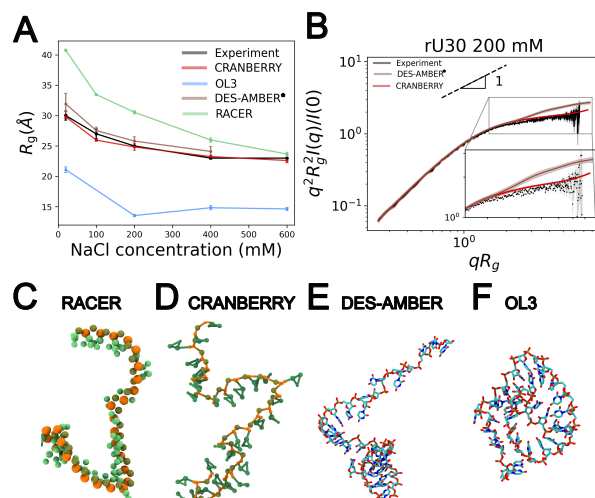


Fig. 5. Radius of gyration and scattering behaviors for disordered RNAs as sampled with different atomistic and coarse-grained force fields in comparison to experiments (A) R_g of rU30 varying salt conditions, comparison between experimental data (93) and various force fields. Error bars are calculated by block averages using the last 3/4 of the trajectory. (B) Dimensionless Kratky plot of rU30 under 200 mM NaCl, showing comparison between experimental data, CRANBERRY, and DES-AMBER; asymptotic rod-limit scaling is indicated by a dashed black line. (C-F) Final configuration of rU30 under 200 mM NaCl using RACER(53), CRANBERRY, OL3 (32) or DES-AMBER (92). (*Note that here we switched to DES-AMBER 3.20 (92) rather than DES-AMBER 2017 (33) as 3.20 improves the predictions on the disordered RNA systems, thus serving as a better benchmark.)

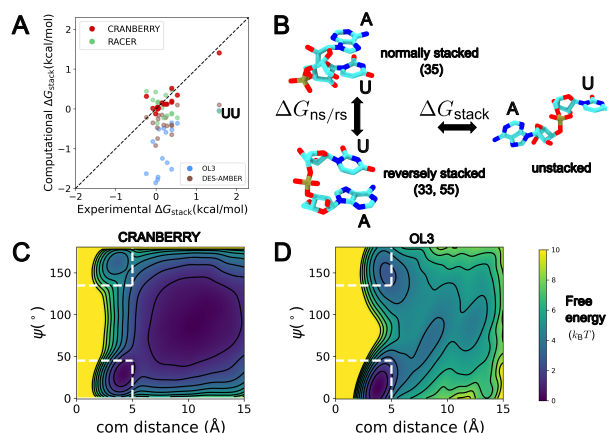


Fig. 6. Stacking free energies. (A) Comparison between experimental (94) and computational ΔG_{stack} . The scatter points with $\Delta G_{\text{stack}}^{\text{exp}} \approx +1.8 \text{ kcal} \cdot \text{mol}^{-1}$ correspond to UU stacking. (B) Snapshots of unstacked, normally stacked, and reversely stacked states (defined in Section Methods: Nonbonded Interactions). (C-D) FES of dinucleotide monophosphate ApU, under CRANBERRY and OL3, respectively. The coordinates are the center of mass distance between two bases and the angle between the normal vectors of the two base planes. The white rectangles denote the regimes of stacked conformations.

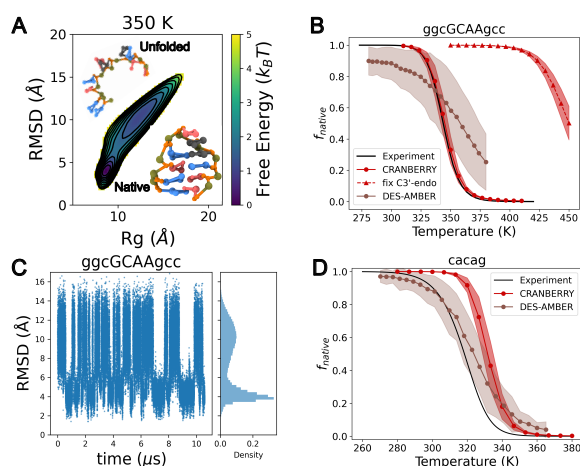


Fig. 7. Melting and *de novo* folding of tetraloops and helices. (A) FES of ggcGCAAgcc at 350 K, with the visualizations of native and unfolded conformations. The contour lines are plotted every 0.5 $k_B T$. (B) Melting curve of ggcGCAAgcc by plotting fraction of native conformations, using DES-AMBER (taken from Ref. (33) by summing up native and intermediate fractions for a consistent definition of the native fraction, see SI Appendix Method for detailed definition of native states) or CRANBERRY, with or without fixing sugar pucker states to C3'-endo. The experimental curve is plotted using the van 't Hoff equation ($K_{eq} = \exp(\Delta H^0 / RT - \Delta S^0 / R)$) and experimental folding enthalpy and entropy. For tetraloop folding $K_{eq} = f / (1 - f)$, and for double-stranded cacag $K_{eq} = 2f / [(1 - f)^2 (c_{ss} / c^\ominus)]$ with the concentration for the single strands denoted as c_{ss} and the standard concentration $c^\ominus = 1$ M. Errors are represented by the shaded area, and are calculated by blocking analysis. (C) *de novo* folding of ggcGCAAgcc showing RMSD over time. (D) Melting curves of the cacag double helix from experiments, DES-AMBER and CRANBERRY.

Stacking free energy. After fine-tuning in our parameterization procedure, CRANBERRY not only predicts stacking free energy at the correct scale, but also captures the correlation of sequences (Fig. 6A, $R^2 = 0.79$). RACER, despite not being optimized against stacking free energy data, does not overstabilize the stacking free energies, except for UU stacking. Both OL3 and DES-AMBER overestimate the stacking free energies of dinucleotide monophosphate by approximately $1 \text{ kcal} \cdot \text{mol}^{-1}$. This is a substantial error given that the system consists of only two nucleotides. This finding is consistent with previous benchmarks (40, 41). Three states emerge in the FES under the coordinates of the center of mass distance and base normal angles ψ : normally stacked, reversely stacked, and unstacked (Fig. 6C and D, and SI Appendix Fig. S9). We note that the experimental stacking free energies are insufficient to fully parameterize the stacking across normally stacked (35), and reversely stacked (33, 55) orientations (Fig. 6B and Section Methods: Nonbonded Interactions for definitions of 33, 35, and 55). Future experimental measurements distinguishing these stacked states, or more accurate all-atom force fields, could aid in parameterization of stacking interactions in the coarse-grained models.

Melting and *de novo* folding of tetraloops and helices.

Although transferability across wide temperature ranges is an inherent problem in coarse-graining (46, 47), CRANBERRY predicts the melting temperatures of ggcGCAAgcc and cacag within 15 K of the experimental values, comparable to the performance of DES-AMBER (33) (Fig. 7B and C). However, the cooperativity (sharpness of the melting

curves) is steeper and closer to the experimental behavior of CRANBERRY (Fig. 7B and C). Consequently, the FES near the melting temperature exhibits cooperative folding features with two distinct free-energy wells (Fig. 7A and SI Appendix Fig. S10). Starting from an unfolded chain, CRANBERRY folds into native tetraloop structures reversibly within $10 \mu\text{s}$ at the experimental melting temperature 344 K (Fig. 7C) with a minimum RMSD of 1.4 Å.

The predicted melting of ggcGCAAgcc is highly cooperative (Fig. 7B) relative to DES-AMBER. This is consistent with our previous experience developing *Upside* 1 and 2 for proteins (70, 72). *Upside* 1 showed weaker cooperativity when the energy function was trained only on native states. Cooperativity improved when we trained using a dual-target loss function (native and disordered states) for *Upside* 2 (72). CRANBERRY's fine-tuning approaches serve as a physical regularizer, providing an effect similar to the dual-target loss function for proteins, and producing cooperative folding behavior for CRANBERRY. The less-cooperative behavior of DES-AMBER may be attributed to residual structures in the unfolded state arising from overstabilized nonbonded interactions in the all-atom force fields (90). Such an overstabilization reduces the enthalpy of disordered states, thereby reducing the magnitude of the folding enthalpies. Indeed, all-atom models of proteins and RNAs, such as DES-AMBER, predict significantly lower folding enthalpies than experiments (33, 95), resulting in their less cooperative unfolding transitions.

The simulated melting temperature of ggcGCAAgcc is increased by as much as ~ 100 K when all sugar pucker states are fixed to C3'-endo while keeping all other settings the same (Fig. 7B). This finding supports the hypothesis that sugar pucker exerts a non-negligible entropic effect on melting behavior. Recently, Zhang (96) proposed a back-of-the-envelope estimation of the entropy loss during the crystallization of polypropylene. The loss of entropy per dihedral is estimated to be $k_B \ln 3$, because each dihedral is fixed to one state from three states when crystallization occurs. This logic can be extended to estimate the effect of sugar pucker on the folding entropy change of ggcGCAAgcc, where the changes in entropy per nucleotide due to sugar pucker could be approximated by $k_B \ln 2$ (unfolded: freely sampling C3'-endo and C2'-endo, folded: restricted to C3'-endo). Using experimental entropy and enthalpy values, the change in melting temperature caused purely by the fixing of sugar pucker is estimated to be $T_m \cdot 10k_B \ln 2 / \Delta S_{exp} \approx 60$ K. This crude estimate agrees on the order of magnitude with our computational result, which underscores the importance of including sugar pucker in CG models to capture thermodynamics and melting behaviors accurately. This is consistent with the theoretical analysis on the temperature transferability of coarse-grained models (47, 97), where including more fine-grained information provides better estimates of the entropy and hence achieves better melting behavior, although the lack of an explicit solvent makes the assignment of the melting entropy a challenge.

Sugar pucker FES. CRANBERRY simulations of an isolated guanosine monophosphate (GMP) have $\sim 1 : 1$ equilibrium between C3'-endo and C2'-endo at 300 K (Fig. 8A). CRANBERRY estimates that the difference in free energy between the C3'-endo and the C2'-endo for a guanosine monophosphate is $\Delta G_{3 \leftrightarrow 2} = G_{C2'-endo} - G_{C3'-endo} \approx +0.2 k_B T$ (C3':C2'=55:45). In GMP NMR experiments, $\Delta G_{3 \leftrightarrow 2}$ is reported from $-0.4 k_B T$ (40:60) (82) to $-0.8 k_B T$ (20:80) (74, 75). Thus, we consider CRANBERRY's performance satisfactory for the thermodynamics of the C3'-endo:C2'-endo transition.

For G7 in the A-form helix (PDB 3ND4), the local minimum of C2'-endo disappears and becomes a shoulder with free energy $4 \sim 5 k_B T$ higher than the C3'-endo (Fig. 8A). We did not tune specifically for this behavior; rather, it emerges for the nucleotides in the double helix after the ConDiv training and fine-tuning. This shift in preference is qualitatively consistent with experiments and all-atom predictions (74, 75, 82), supporting our modeling approach for sugar pucker. Using an all-atom force field, the free energy difference $\Delta G_{3 \leftrightarrow 2}$ is predicted to be about $+10 k_B T$ and the barrier is estimated to be about $20 k_B T$ (75). The qualitative agreement between CRANBERRY and the all-atom simulations can be explained by the following mechanisms for C2'-endo destabilization in A-form helices. The destabilization of the C2'-endo has been attributed to the disruption of base stacking by the C2'-endo form (75). CRANBERRY captures this effect via sugar pucker and base-stacking energy functions. Furthermore, steric effects of 2'-OH and hydration effects have also been proposed to destabilize C2'-endo (73, 75, 98). These effects are not yet fully captured by CRANBERRY, which may explain the lack of quantitative agreement with the all-atom predictions (75). However, to the best of our knowledge, no direct experimental measurements exist for the free-energy difference or the height of the barrier. Along the double helix, CRANBERRY predicts that the fraction of north conformations remains at 94–96% (Fig. 8C). ("North" includes C3'-endo and nearby states; we report north fractions to capture contributions from conformations near C3'-endo, e.g., C2'-exo, C4'-exo.) This is consistent with observations that in A-form RNA helices virtually all helical nucleotides are C3'-endo, whereas C2'-endo is observed mainly outside the helical region (e.g., bulges, loops, termini) (73, 75, 99, 100).

Limitations. Similarly to other CG models (53, 62), CRANBERRY treats electrostatics at the level of the Debye-Hückel approximation, which is strictly valid only at low salt concentrations. Although CRANBERRY captures salt-dependent size changes in disordered RNA, its ability to model dynamics at higher salt concentrations remains unclear. In the future, explicit ion models may need to be developed, as demonstrated in other CG models (5, 60, 101).

Another limitation is that CRANBERRY currently can not model divalent ions such as magnesium or base-ion interactions. The only electrostatic interactions currently explicitly included are repulsions between the phosphates. CRANBERRY is not expected to model ion-dependent structures

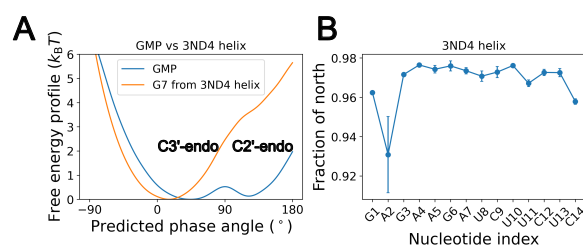


Fig. 8. Sugar pucker FES and dynamics. (A) FES of the predicted phase angle of a guanosine monophosphate (GMP) and the G7 in a 3ND4 helix. (Negative phase angles ($-90^\circ - 0^\circ$) are shifted from the range ($270^\circ - 360^\circ$)) (B) Equilibrium fraction of north conformation of each nucleotide over the nucleotide indices. (The north population is defined as having a phase angle in the range $0^\circ - 90^\circ$ or $270^\circ - 360^\circ$, which are usually dominated by C3'-endo, see definitions in Section Results and Discussion: Sugar Puckering Motion and Fig. 2A). Error bars are calculated using the last 3/4 of the trajectory. The terminal nucleotides A0 and U15 are not included.

such as G-quadruplexes. In the future, CRANBERRY could be extended to include these terms.

CRANBERRY can qualitatively capture sugar pucker. In helical regions, CRANBERRY does not destabilize south conformations (or mostly, C2'-endo) enough compared to all-atom models. This is primarily due to the absence of explicit 2'-OH and backbone sterics in the coarse-grained model. This effect could be addressed by coupling the $\eta - \theta$ map with sugar pucker, favoring the C3'-endo in the helical regimes of the $\eta - \theta$ map (78, 102, 103). There are also known correlations between the map $\eta - \theta$ and sugar pucker states (78), which could be included as well. However, these are considered second-order effects and are omitted in our initial CRANBERRY implementation. Introducing the $\eta - \theta$ map might also help stabilize various turn motifs in RNA.

CRANBERRY currently models only the 25 most frequent base-pairing geometries, which together account for 88% of cases in the fragment data set (Materials and Methods: Data Set). We note that our approaches are entirely compatible with new experimental data or more accurate *ab initio* calculations. In the future, all 156 known base-pairing geometries could be parameterized and incorporated.

CONCLUSIONS

We highlight the following design elements that enable CRANBERRY's performance: contrastive divergence training (70), the fine-tuning process that greatly improves the disordered states and folding cooperativity, and the incorporation of physically-motivated terms such as non-canonical base pairing and sugar pucker.

CRANBERRY's major accomplishments are:

1. Stabilizes helices, tetraloops, and pseudoknots at room temperature within 2–6 Å RMSD of the crystal structures;
2. Captures the salt-dependent expansion of disordered rU30 and reproduces SAXS profiles at physiological ionic strengths;

3. Predicts dinucleotide stacking free energies to within $0.5 \text{ kcal mol}^{-1}$ of experiment, avoiding the systematic over-stabilization common in all-atom force fields;
4. Recovers cooperative melting of the tetraloop ggcG-CAAgcc and double helix cacag within 15 K of experiments but exhibits better cooperativity compared to all-atom;
5. Folds ggcGCAAgcc *de novo*, reversibly from an extended strand with minimum RMSD as $1.4 \text{ \AA}$;
6. Reproduces sugar-pucker preferences in mononucleotides and helices.

We developed CRANBERRY, a six-site coarse-grained model for RNA that simultaneously captures sugar-pucker dynamics, noncanonical base pairing, native and disordered states, and thermodynamics. Employing a linear-regression-based many-body bonded potential for sugar puckering transitions, CRANBERRY naturally reproduces both the fast kinetics and the equilibrium populations of ribose puckers. Our contrastive-divergence training, augmented with fine-tuning against experimental stacking free energies and disordered-RNA Rgs, produces nonbonded potentials that balance native-state fluctuations with realistic melting behavior. CRANBERRY's computational efficiency opens the door to routine exploration of RNA folding landscapes and large-scale ribonucleoprotein dynamics. We anticipate that CRANBERRY will serve as a practical tool for simulating complex RNA assemblies.

Materials and Methods

Data Sets. To parameterize the model, we curated two data sets: a fragment and a ConDiv data set. The fragment data set was used to parameterize bonded potentials and provide initial parameters for base stacking and pairing potentials. The ConDiv data set was used to train base stacking, pairing, and spline potentials via ConDiv (67–70).

The fragment data set was constructed by filtering the non-redundant 3D RNA structure data sets (104), removing structures with a resolution higher than $3 \text{ \AA}$. This data set includes the RNA components of the RNA-protein and RNA-ligand complexes (e.g., ribosomes and riboswitches). This data set was then used to compute various statistics: 1) coarse-grained degrees of freedom (bonds, angles, and dihedral angles) and phase angles (via the Cremer and Pople method (80)), used to parameterize the linear regression model for predicting sugar puckering, as discussed in Section Methods: Sugar Puckering Energy Function; 2) histograms of bonds, angles, and dihedrals used for parameterizing bonded potentials; 3) histogram of degrees of freedom of base pairing ($r, \theta_1, \theta_2, \phi_1, \phi_2, \psi$ (Fig. 1b)), used to create initial parameters for base pairing before ConDiv training. Base pairs are identified by DSSR (87). This data set contains 1150 RNAs, 127,293 nucleotides, and 162,333 valid base pairs. It is usually used for local features and should be regarded as a fragment library. As a result, we did not filter chain breaks or perform MD simulations on this data set.

The ConDiv data set was constructed by further filtering the non-redundant list using the NAKB (Nucleic Acid Knowledgebase) webserver (105). This data set includes RNA-only structures (crystal structures, Cryo-EM, or NMR) composed exclusively of standard nucleotides. The molecular weight range is restricted to 6.6–33 kDa, which corresponds to roughly 20–100 nucleotides. This range was chosen to ensure that the trajectories remain computationally feasible during training. We further excluded quadruplexes (which require modeling explicit base-ion interactions) and structures containing ligands. We also removed any structure that contains chain breaks. In addition, we applied a $3 \text{ \AA}$ resolution cutoff for X-ray and Cryo-EM structures. In the end, this data set contains 64 X-ray or Cryo-EM structures and 203 NMR structures.

Bonded Energy Functions. The bonded potential, U_b , includes bond (U_{bond}), angle (U_{angle}), and dihedral (U_{dihedral}). The bond and angle terms are harmonic. The dihedral term uses the combined bending-torsion potential (CBT) (106), which mitigates numerical instabilities when three atoms become colinear and the dihedral becomes degenerate. Several approaches exist to address dihedral instabilities (106–108); we selected CBT for its simplicity.

$$\begin{aligned}
 U_b &= U_{\text{bond}} + U_{\text{angle}} + U_{\text{dihedral}} \\
 &= \sum_{i \in \text{bonds}_{\text{nsp}}} k_b (r_i - r_{0,i})^2 \\
 &\quad + \sum_{i \in \text{angles}_{\text{nsp}}} k_\theta (\theta_i - \theta_{0,i})^2 \\
 &\quad + \sum_{i \in \text{dihedrals}_{\text{nsp}}} k_\phi \sin^3 \theta_{i-1} \sin^3 \theta_i (1 + \cos(\phi_i - \phi_{0,i})),
 \end{aligned} \tag{3}$$

where θ_{i-1}, θ_i are the angles formed by 3 of the 4 particles corresponding to the dihedrals (angle 1-2-3 or 2-3-4 when the dihedral is formed by 1-2-3-4), and the factors $\sin^3 \theta_i$ are included to prevent diverging forces as internal angles approach 180° (106). Bonds, angles, and dihedrals not related to sugar puckering are denoted $\text{bonds}_{\text{nsp}}, \text{angles}_{\text{nsp}}, \text{dihedrals}_{\text{nsp}}$ (where “nsp” stands for “not sugar pucker”). The parameters are listed in Table S1-3 of the SI Appendix.

Sugar Puckering Energy Function. The sugar puckering contribution, U_{pucker} , is represented by a many-body bonded potential. This many-body bonded potential interpolates between bonded potentials parameterized for C3'-endo U_3 and C2'-endo U_2 . U_3 and U_2 are standard bond, angle, and dihedral potentials, using the same formulas as $U_{\text{bond}}, U_{\text{angle}}, U_{\text{dihedral}}$ in Eq. 3. The difference lies in the sets of bonds, angles, and dihedrals to which they are applied. This choice is because some coarse-grained degrees of freedom strongly correlate with sugar puckering states (Fig. 2D). The parameters for these bond/angle/dihedrals are listed in

Table S4-6 of the SI Appendix .

$$U_{\text{pucker}} = \sum_{j \in \text{nucleotides}} \left\{ \sigma(P_j; P_0, a_s) (U_2 + \Delta U_{3 \leftrightarrow 2}) + [1 - \sigma(P_j; P_0, a_s)] U_3 \right\} \left[1 - k_g \text{Gaussian}(P_j; P_0, a_g) \right],$$

$$\sigma(P_j; P_0, a_s) = \frac{1}{1 + \exp[-a_s(P_j - P_0)]},$$

$$\text{Gaussian}(P_j; P_0, a_g) = \exp[-a_g(P_j - P_0)^2], \quad (4)$$

where $\sigma(P_j, P_0, a_s)$ serves as a switching function that transitions the bonded potential between the C3'-endo and the C2'-endo. $\text{Gaussian}(P_j, P_0, a_g)$ controls the energy barrier between C3'-endo and C2'-endo. P_j is the predicted phase angle for the sugar in the nucleotide j as defined in Eq. 2 in Section Results and discussion: Sugar Puckering Motion, where $P_0 = 90^\circ$ is the switching threshold, and $a_s = (0.085^\circ)^{-1}$, $a_g = (4.38 \times 10^{-6} \text{ }^\circ)^{-2}$ control the sharpness of the switching function and the Gaussian functions, respectively (Fig. 2E). $\Delta U_{3 \leftrightarrow 2} = 130 \text{ kJ mol}^{-1}$ is parameterized by the population ratio for free nucleotides. $k_g = 0.947$ controls the overall strength of the Gaussian modification part. This sugar pucker potential is applied to each nucleotide independently. For simplicity, if the 5'-phosphate is missing, the terminal nucleotide is modeled by the bonded terms as in C3'-endo.

Nonbonded Interactions. The nonbonded potential, U_{nb} , includes excluded volume interactions (U_{wca}), base stacking (U_{stacking}), base pairing (U_{pairing}), electrostatics (U_{elec}), and spline-based interactions between backbone (phosphate P and sugar sites S2 and S3) and base (U_{spline}).

$$U_{\text{nb}} = U_{\text{wca}} + U_{\text{stacking}} + U_{\text{pairing}} + U_{\text{elec}} + U_{\text{spline}}. \quad (5)$$

The excluded volume interactions are represented by the Weeks–Chandler–Andersen (WCA) potential (109).

$$U_{\text{wca}} = \sum_{ij} \begin{cases} \varepsilon_{\text{wca}} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \varepsilon_{\text{wca}} & r < r_{\text{cut}}, \\ 0 & r \geq r_{\text{cut}}, \end{cases} \quad (6)$$

where the energy parameter ε_{wca} is globally set to be 5 kJ mol^{-1} , and $\sigma_{ij} = (\sigma_i + \sigma_j)/2$ is the average diameter of the site. r_{ij} denotes the distance between particles. The σ_i parameters are listed in SI Appendix Table S8. Note that, in addition to the standard sites, the BC virtual site also participates in WCA interactions.

The base stacking potential is anisotropic and involves one radial switching function and three angular switching func-

tions between two bases.

$$U_{\text{stacking}} = \sum_{i,j,\text{type}} U_{ij}^{\text{s,type}} f(r_{ij}; r_{ij}^{\text{s,type}}, \Delta r_{ij}^{\text{s,type}}, a_r^{\text{s}}) \times g(\cos \theta_{ij,1}; \cos \theta_{ij,1}^{\text{s,type}}, a_\theta^{\text{s}}) \times g(\cos \theta_{ij,2}; \cos \theta_{ij,2}^{\text{s,type}}, a_\theta^{\text{s}}) \times g(\cos \psi_{ij}; \cos \psi_{ij}^{\text{s,type}}, a_\psi^{\text{s}}),$$

$$f(r; r_0, \Delta r, a) = \frac{1}{2} \tanh[a(|r - r_0| - \Delta r)] - \frac{1}{2},$$

$$g(x; y, a) = \frac{1}{2} \tanh[a(x - y)] + \frac{1}{2},$$

(7)

where i and j are the indices of the nucleotides. Visual representations of r_{ij} , θ_1 , and θ_2 are shown in Fig. 1b, where $\cos \theta_{ij,1} = \hat{r}_{ij} \cdot \hat{z}_i$, $\cos \theta_{ij,2} = \hat{r}_{ij} \cdot \hat{z}_j$. r_{ij} is the distance between BC of the nucleotide i and that of nucleotide j , and $\hat{r}_{ij}$ is its unit vector. $\hat{z}_i$ and $\hat{z}_j$ are unit vectors along the z-axes of bases i and j , respectively. Possible types are 33, 55, and 35, corresponding to interactions via 3'-faces and 5'-faces. The 5'-face is defined as the side of the base plane containing the atom C5'. In canonical A-form helices, neighboring bases stack via 3'-5' faces (35), and this type of stacking is termed 'normally stacked'. Accordingly, 33 and 55 types of stacking are denoted as 'reversely stacked' (Fig. 6b). All the parameters are provided in SI Appendix Table S9. The experimental enthalpies (94) are asymmetric with respect to the swapping the first and second nucleotide type. For example, the enthalpy of UG is different from that of GU. Thus, we allow the interaction strength matrix for the 35 stacking $U_{ij}^{\text{s,35}}$ to be asymmetric. In contrast, the matrices for the 33 stacking $U_{ij}^{\text{s,33}}$ and 55 stacking $U_{ij}^{\text{s,55}}$ are symmetric by definition. The base pairing potential is modeled as an anisotropic interaction with one radial component and five angular components.

$$U_{\text{pairing}} = \sum_{i,j,\text{type}} U_{ij}^{\text{p,type}} f(r_{ij}; r_{ij}^{\text{p,type}}, \Delta r_{ij}^{\text{p,type}}, a_r^{\text{p}}) \times h(\theta_{ij,1}; \theta_{ij,1}^{\text{p,type}}, \Delta \theta_{ij}^{\text{p,type}}, a_\theta^{\text{p}}) \times h(\theta_{ij,2}; \theta_{ij,2}^{\text{p,type}}, \Delta \theta_{ij}^{\text{p,type}}, a_\theta^{\text{p}}) \times h(\phi_{ij,1}; \phi_{ij,1}^{\text{p,type,1}}, \Delta \phi_{ij}^{\text{p,type,1}}, a_\phi^{\text{p}}) \times h(\phi_{ij,2}; \phi_{ij,2}^{\text{p,type,2}}, \Delta \phi_{ij}^{\text{p,type,2}}, a_\phi^{\text{p}}) \times g(\cos \psi; \cos \psi_{ij}^{\text{p,type}}, a_\psi^{\text{p}}),$$

$$\phi_{ij,1} = \text{atan2}(\hat{r}_{ij,1}^\perp \cdot \hat{y}_i, \hat{r}_{ij,1}^\perp \cdot \hat{x}_i),$$

$$\phi_{ij,2} = \text{atan2}(\hat{r}_{ji,2}^\perp \cdot \hat{y}_j, \hat{r}_{ji,2}^\perp \cdot \hat{x}_j),$$

$$\hat{r}_{ij,1}^\perp = \hat{r}_{ij} - (\hat{r}_{ij} \cdot \hat{z}_i) \hat{z}_i,$$

$$\hat{r}_{ji,2}^\perp = \hat{r}_{ji} - (\hat{r}_{ji} \cdot \hat{z}_j) \hat{z}_j,$$

$$h(\theta; \theta_0, \Delta \theta, a) = \frac{1}{2} \tanh[a(\cos|\theta - \theta_0| - \cos \Delta \theta)] - \frac{1}{2}, \quad (8)$$

where $\hat{x}_i, \hat{y}_i, \hat{z}_i$ are the unit vectors along the local x/y/z-axis of each base (Fig. 1). $r_{ij}, \theta_{ij,1}, \phi_{ij,1}$ are the BC coordinates of base j in the local spherical coordinate system of base i , with $\hat{r}_{ij,1}^\perp$ being the projection of $\hat{r}_{ij}$ onto the plane of base i (Fig. 1b). Their frequencies are shown in SI Appendix Fig. S1. These 25 geometries are denoted by “type” in Eq. 8. The corresponding parameters are tabulated in SI Appendix Table S10. Base pairing between adjacent bases on the same strand is explicitly disallowed.

Electrostatic interactions are approximated using a Debye-Hückel type screened potential to account for salt effects, following the 3SPN.2 CG model for DNA (62)

$$U_{\text{elec}} = \sum_{i,j} \frac{q_i q_j \exp[-r_{ij}/\lambda_D]}{4\pi\epsilon_0\epsilon(T,C)r_{ij}},$$

$$\lambda_D = \sqrt{\frac{\epsilon_0\epsilon(T,C)}{2\beta N_A e_c^2 I}},$$

$$\epsilon(T,C) = \epsilon(T)a(C),$$

$$\epsilon(T) = 249.4 - 0.788T/K + 7.20 \times 10^{-4}(T/K)^2,$$

$$a(C) = 1.000 - 2.551C/M + 5.151 \times 10^{-2}(C/M)^2 - 6.889 \times 10^{-3}(C/M)^3, \quad (9)$$

where q_i and q_j are the charges on phosphate i and j , r_{ij} is the inter-site distance, λ_D is the Debye screening length, ϵ_0 is the dielectric permittivity of vacuum, $\epsilon(T,C)$ is the dielectric permittivity of the solution. β is the inverse thermal energy $(k_B T)^{-1}$, N_A is Avogadro’s number, e_c is the elementary charge, and I is the ionic strength of the solution. The dielectric permittivity of the solution $\epsilon(T,C)$ is a function of temperature T and the concentration of the NaCl solution C . As in the CG model 3SPN.2 (62), the charge of the phosphate site is reduced to -0.6 to approximate the effect of Manning counter-ion condensation (109–111).

The spline potential provides a flexible way to represent the complex interactions of phosphate–phosphate, sugar–sugar, phosphate–sugar, phosphate–base (112), and sugar–base.

$$U_{\text{spline}} = \sum_{i,j} h(r_{ij}), \quad (10)$$

where $h(r)$ is represented by natural cubic splines. For each pair of residue types, different spline functions are trained (Table S11 and Fig. S2 of the SI Appendix). The spline functions between base sites are set to zero, since their interactions are already captured by base-stacking and base-pairing. For all the nonbonded interactions, pairs of sites separated by three or fewer bonds are excluded. CRANBERRY includes approximately 1000 parameters in total.

Langevin dynamics. Following 3SPN.2 (62), Langevin dynamics is used to propagate the dynamics of the system with a time step of 5 fs. To assign the damping constant, the Einstein relation is used

$$D = \frac{1}{\beta\xi_m} = \frac{1}{\beta\sum_i \xi_i} = \frac{1}{\beta\gamma\sum_i m_i}, \quad (11)$$

where ξ_m is the molecular friction coefficient, and $\xi_m = \sum_i \xi_i$ where ξ_i is the friction coefficient of each site, and i is summing all the sites in the molecule. We ignore the difference in the damping coefficients between sites with different masses. We use an empirical scaling relation (113) to account for RNA length in diffusivity $D = 4.58 \times 10^{-10} N^{-0.39} \text{m}^2 \text{s}^{-1}$, where N is the number of nucleotides.

Parametrization approach. Using the fragment data set, we manually optimized the bonded potential to correctly capture the data set’s distribution of bonded degrees of freedom (SI Appendix Fig. S3). Then, we determine the steric repulsion parameters by two criteria. First, we compared the peak positions of the radial distribution function of all sites in all-atom simulations of an A-form helix in water with the sum of LJ diameter (2.8 Å) for the water and the diameter of the coarse-grained sites. Secondly, we ensure that for the canonical A-form helix 3ND4, the steric repulsion is minimal. Afterward, we initialize the base pairing and base stacking interactions with parameters derived from the fragment data set. We use all-zero values as initial parameters for spline interactions, since they are minor contributions to the total energy of the system. Following the 3SPN.2 parameterization protocols (62), the strength parameters for the stacking are initialized by scaling the experimental enthalpies (94).

After obtaining the initial parameters, we performed the ConDiv training (70) using the ConDiv data set of 267 RNAs. We approximate the reference distribution of these RNAs at room temperature via restrained simulations near their structures. Restrained simulations are performed by randomly selecting $3N_{\text{sites}}$ native contacts and then applying harmonic restraints to these contacts (70, 72). At each epoch, we run RNA trajectories in the training set for around 1 ns, and we calculate approximate derivatives of the Kullback-Leibler divergence between the reference distribution and the distribution produced by CRANBERRY (70). The Adam optimizer (114) is used for optimization. The ConDiv method is applied only to the nonbonded base pairing, stacking, and spline interactions, while the bonded terms are inferred from the fragment data set. We run 30 epochs to achieve convergence. We further fine-tune the stacking parameters by running benchmark simulations of di-nucleotide monophosphate for which the experimental stacking free energies are available (94). In this fine-tuning step, we only tune the strength parameter of the stacking potential $U_{ij}^{\text{s,type}}$, for the 35, 55, and 33 classes of stacking geometries. We found that the experimental stacking free energies ΔG_{stack} do not distinguish between normally and reversely stacked conformations. As a result, we use a hybrid scheme for this matching process, where we simultaneously match the experimental stacking free energies ΔG_{stack} (Eq. 2 of the SI Appendix) and the free energy difference between normally and reversely stacked $\Delta G_{\text{ns/rs}}$ (defined in Eq. 3 of the SI Appendix) estimated from OL3. Due to limited data, we assume that the strength of the 55 types to be the same as for the 33 types. The matching process is carried out by performing the secant method (115). Furthermore, we globally scaled

the spline potential (backbone-backbone and backbone-side chain interactions) so that the Rg of rU30 matches the experimental value at 200 mM salt. We note that due to the limit of the training set size and information about dynamical ensembles, the ConDiv method tends to overfit, such that the structures might be overstabilized. This behavior is similar to the overstabilization tendency of all-atom force fields when the main goal is to stabilize different known DNA/RNA structures (31–33, 40, 92, 116, 117). Similar behavior is also observed for all-atom force fields (90) and coarse-grained models (70, 72) developed for proteins. A similar issue has recently been resolved by introducing regularization terms for protein binding affinities (118). Thus, this underscores the importance of including disordered RNA and thermodynamic fine-tuning, because they also serve as physical regularizers for the model.

MD Simulation Protocols and FES Calculations. The simulations of all-atom and coarse-grained force fields follow common practices (33, 40, 41, 53), with details of simulation setup and calculations of stacking, native fraction, and SAXS profile included in the SI Appendix.

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